



**Theoretical Biology and Biophysics (T-6)
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Wellcome Leap Resistance Networks Program

**Subject: Universal Properties of Plasmid Evolution and Robust Risk Forecasting
for Preemptive pARG Intervention**

Dear Members of the Review Committee:

I am writing in support of the above proposal, led by Dr. Philip J. Gerrish.

My colleagues and I have developed methods for estimating epidemic growth rates and reproductive numbers from incomplete and noisy incidence data. In Ke et al. (2021), we estimated the early exponential growth rate and basic reproductive number of SARS-CoV-2 by fitting a mechanistic SEIR-type model jointly to case and death data. The analysis accounted for reporting delays, incomplete detection, variation in latent and infectious periods, and uncertainty in the relation between epidemic growth rate and R . It also demonstrated how an estimated reproductive number can be used to quantify the strength and timing of interventions required to control an outbreak.

In subsequent work with Trejo et al. (2024), we developed a Bayesian, nonparametric method for estimating a time-varying reproductive rate and the corresponding effective reproduction number, $R(t)$, from incidence time series. The method separates the population-level transmission process from the temporal profile of infectiousness following an infection. It provides posterior uncertainty intervals and posterior predictive distributions and can be used to compare the output of a mechanistic epidemic model with a less model-dependent estimate derived directly from observations.

The same structure is applicable to the spread of antibiotic-resistance plasmids, although several biological extensions will be required. A susceptible unit can be defined as a bacterial cell, host population, or sampled site that does not carry the plasmid. Acquisition of the plasmid then constitutes an incidence event, and a plasmid-bearing cell or population becomes a potential source of subsequent acquisition events. The infectiousness profile is replaced by a plasmid-specific transfer and persistence profile describing how the capacity to generate new plasmid acquisitions changes with time or

generations after acquisition. The inferred reproductive rate would capture changes in donor-recipient contact, antibiotic exposure, host and plasmid composition, conjugation efficiency, and environmental conditions.

This formulation would provide a time-dependent plasmid reproduction number with a directly interpretable threshold: values above one indicate conditions under which plasmid carriage is expected to expand, whereas values below one indicate contraction. It would also permit detection of transmission waves associated with antibiotic exposure or evolutionary change. Importantly, the Bayesian estimate could serve as an empirical standard against which the project's mechanistic models of plasmid-host coevolution are evaluated. Plasmid loss, vertical clonal expansion, population structure, and incomplete detection should be represented explicitly rather than treated as direct analogues of recovery and case reporting in a human epidemic. The proposal recognizes these distinctions and provides an appropriate setting in which to develop and test the necessary extensions.

I support this proposal. If funded, I would be willing to advise the project on the definition and estimation of plasmid reproductive numbers and on validation of its mechanistic models against longitudinal data.

Sincerely,
Nick Hengartner
Los Alamos National Laboratory
Nick Hengartner
Scientist